

A QR Decomposition Based RLS Algorithm with Forgetting Factor for Adaptation of AR EEG Features

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Abstract— Electroencephalography (EEG) is an effective modality being used to develop brain computer interface (BCI) systems, but the EEG signal is predominantly marked with low signal-to-noise ratio. Thus, extracting information of interest from EEG data becomes quite a challenging task. Brain activity is usually modeled as an adaptive autoregressive (AAR) model and various adaptive algorithms are implemented to obtain its parameters. Here, QR decomposition based recursive least squares (QR-RLS) algorithm with forgetting factor is derived for AAR parameter estimation and realized to track the EEG features with numerical stability. EEG features from motor imagery datasets are extracted by the proposed method and classified with Linear Discriminate Analysis. For further validation, a comparative analysis is obtained between the presented methodology and already existing EEG processing methods, i.e., Least Mean Squares, RLS and conventional QR-RLS. Results indicate that EEG features extracted from the proposed algorithm provide better classification accuracy than other adaptive algorithms. Thus, promising towards development of reliable and proficient brain computer interface systems.

Keywords—Brain Computer Interface Systems (BCIs); Adaptive Algorithms.; Least Mean Square (LMS), Recursive Least Squares (RLS); QR-RLS Algorithm.

I. INTRODUCTION

Present era is being marked as an era of mapping between various technologies and fields. Biomedical Engineering happens to be one of the fastest growing areas in this aspect, thus leading towards development of numerous biomedical modalities such as electroencephalogram (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI) and functional near-infrared spectroscopy (fNIRS) [1]. Among these modalities, EEG based research is being preferred extensively due to its high temporal resolution, low cost and easy portability [1], [2]. On the other hand, fMRI, MEG and fNIRS require comparatively more expensive and bulkier instruments [3]. Higher temporal resolution of EEG benefits in recording and decoding complex patterns of brain signals within fractions of a second. Thus, EEG applications find potential usage in tracking of high rate brain activity. However, the limited spatial resolution [1], [3] can deplete their performance but appropriate selection of electrode sensing

positions and BCI controls can help in overcoming this drawback [4].

EEG based brain-computer interfaces (BCIs) find diverse clinical applications. They have become an emergent communication gateway, capable of substituting the functioning of impaired motor functions. Not only the medical specialists use EEG to detect different disorders of human brain such as epilepsy, memory loss, and sleep disorders [5], but also in diagnostic imaging for identification of damaged brain areas after head injuries and brain tumors. BCI based devices such as wheelchair control by brain thoughts [6] and prosthetic limbs [2] have been developed by researchers to assist the disables. Advancements in various social applications including biometric sensing such as a continuous brain monitoring of driver's and pilot's attention to avoid any accidental situations and also in military applications such as distant supervision of robots and drones in battle fields. In addition to this, BCI based smart homing and smart gaming applications are also creating a promising pathway for BCI researches [2].

BCI applications are still in development stage resulting in the fact that no standardized brain signal acquiring techniques are available [7]. However, EEG signals are normally acquired based on certain experimental paradigms in most existing BCI systems; such as, steady state visual evoked potentials (VEPs), P300 VEPs, auditory evoked potentials, mental tasks evoked potentials, motor imagery potentials and motor movement potentials [8]. The choice of most suitable method for signal acquisition is decided explicitly on the basis of design application and experimental protocol [9].

While devising a BCI system, there are two critical stages of signal processing; the first one is to model brain activity and the second one is to estimate the model's parameters. Researchers have been working for exploiting different types of brain functionality models along with various parameter estimators. BCI systems estimate EEG activity by implementing various adaptive algorithms for online processing. Brain activity is modelled as an adaptive autoregressive (AAR) model [10], [11]. Selection of AAR is based on the fact that it lends to an efficient modeling of the EEG signals as filtered white noise

with some specific energy bands and the EEG time series is considered to be fitted well with an AAR model [11]. This paper aims to contribute towards a numerically stable and optimal algorithm for processing of EEG data and extracting features for motor imagery signals and their proficient decoding to develop a reliable BCI system.

The most promising adaptive algorithms for online EEG signal processing and parameter estimation are least mean squares (LMS), recurring least squares (RLS) and conventional QR-RLS. A comparative study states that LMS, being the simplest, requires more number of iterations for optimal output than RLS. Thus, RLS algorithm has faster convergence speed than LMS [12] and QR-RLS algorithm is proven to be more numerically stable than the conventional RLS [13]. In this paper, QR-RLS with forgetting factor is proposed to extract EEG features. The features obtained from adaptive algorithms are classified based on linear discriminate analysis (LDA). In order to evaluate the performance of QR-RLS with forgetting factor methodology, a comparative analysis was performed. The validation of proposed QR-RLS algorithm against LMS, RLS and conventional QR-RLS was carried out on a series of online motor (hands) movement imagination experiments from [3] and a publicly available movement imaginary dataset [4]. Both data are acquired by utilizing the same experimental protocol. Comparative results depict that the QR-RLS with forgetting factor algorithm is potentially more proficient for parameter characterization than other adaptive algorithms.

II. METHODOLOGY

A. Brain Activation Model

Many relations have been presented to model EEG signals. Out of which, the most commonly used mathematical models are autoregressive (AAR) and autoregressive moving average (ARMA). These parametric methods are considered to be very powerful estimators because of the fact that AAR and ARMA can completely describe the second order statistics of a time varying process with a minimum number of parameters. Here, an AAR model of brain has been used, based on the following equation [14]

$$y(k) = \beta_1(k)y(k-1) + \dots + \beta_p(k)y(k-p) + e(k), \quad (1)$$

where $y(k)$ denotes the value of the EEG electrode signal measured at the time instant k , $\beta_1(k), \dots, \beta_p(k)$ are the coefficients values that are adapted at k^{th} instance of the previous p samples from the EEG signals. The signals are denoted as $y(k-1), \dots, y(k-p)$, at the specific instances respectively. The order of AAR model depends upon value of parameter p . Defining the autoregressive parameters of AAR model and past p samples of EEG data time series in the vector form as

$$\beta(k) = [\beta_1(k) \ \beta_2(k) \ \beta_3(k) \dots \beta_p(k)]^T, \quad (2)$$

and

$$Y(k) = [y(k) \ y(k-1) \ y(k-2) \dots y(k-p)]^T. \quad (3)$$

We obtained for (1)

$$y(k) = Y^T(k)\beta(k) + e(k). \quad (4)$$

Thus, the error in parameter estimation can be written as

$$e(k) = y(k) - Y^T(k)\beta(k), \quad (5)$$

where $e(k)$ indicates the output error of the process for the autoregressive model presented in (1).

B. Parameter Estimation Algorithms

The most promising available adaptive algorithms for online EEG signal processing are least mean squares (LMS), recursive least squares (RLS) and numerically stable variant of standard RLS known as QR-RLS. A brief description of the above mentioned algorithms used for regressing the parameters of (1), is presented here.

1). *Least Mean Squares Estimation.* In LMS algorithm, the coefficients are tuned from sample to sample so that it can minimize mean square error [15]. The LMS algorithm follows following mathematical equation for estimation of EEG parameters [16].

$$\beta(k) = \beta(k) + \frac{uc}{msy} e(k) Y(k), \quad (6)$$

where uc denotes the update coefficient and msy denotes the total variance or power of the brain signal. These parameters are initialized based upon the acquired EEG data properties.

2). *Recursive Least Squares Estimation.* RLS algorithm is described by the following equations [12]

$$S(k) = (1 + uc) P(k-1) Y(k), \quad (7)$$

$$H(k) = S(k) / [1 + Y^T(k) S(k)], \quad (8)$$

$$\beta(k) = \beta(k-1) + e(k) H(k), \quad (9)$$

$$P(k) = P(k-1)(1 + uc) - H(k) S^T(k), \quad (10)$$

where P is inverse of the correlation matrix and H is the gain. The values chosen for model order and update coefficient are of $p=6$ and $uc = 0$, respectively.

3). *QR-RLS with Forgetting Factor Estimation.* The proposed QR-RLS algorithm with forgetting factor can be derived by the following equations. The forgetting factor reduces the influence of old data, i.e., the data obtained from p previous values and hence helps in better estimation based on current data. In least squares parameter estimation, the squared magnitude error $\varepsilon(k)$ for (5) is calculated as [15]

$$\varepsilon(k) = \sum_{i=0}^n w^{k-i} |e(k)|^2, \quad (11)$$

where w is the ‘forgetting/weighting factor’ having a value between zero and one. Using (5) and (11), we obtained

$$\varepsilon(k) = \sum_{i=0}^n w^{k-i} |y(k) - Y^T(k)\beta(k)|^2, \quad (12)$$

in according to the QR decomposition approach proposed by [16], the extended EEG data matrix can be written in the form $Y(k) = [y(k) \quad \sqrt{w}y(k-1) \quad \dots \sqrt{w}^{k-i}y(k-p)]^T$. The solution to this well-known least squares problem is given by [17]

$$P(k)\beta(k) = \Psi(k), \quad (13)$$

$$\text{with } P(k) = \sum_{i=0}^n w^{k-i} Y(k)Y^T(k), \quad \Psi(k) = \sum_{i=0}^n w^{k-i} Y^T(k)y(k),$$

having recursive relations of form

$$\Psi(k) = w\Psi(k-1) - y(k)Y^T(k), \quad (14)$$

and

$$P(k) = wP(k-1) - Y(k)Y^T(k), \quad (15)$$

respectively. Multiplying (13) by $P^T(k)$ from left, we obtained

$$P^T(k)P(k)\beta(k) = P^T(k)\Psi(k), \quad (16)$$

where P is the inverse of auto-correlation matrix. QR decomposition of auto-correlation matrix is obtained by the modified Gram-Schmidt Orthogonalization algorithm, such that $P = QR$, thus (16) becomes

$$[QR]^T[QR]\beta(k) = [QR]^T\Psi(k), \quad (17)$$

where Q is the orthonormal matrix and R is the upper triangular matrix. By simple mathematics, (17) is transformed to

$$R^T Q^T QR\beta(k) = R^T Q^T \Psi(k), \quad (18)$$

Using the matrix identity $Q^T Q = I$ and by simplification, we get

$$R^T R\beta(k) = R^T Q^T \Psi(k), \quad (19)$$

Thus, the parameters $\beta(k)$ can be calculated as

$$\beta(k) = R^{-1} Q^T \Psi(k). \quad (20)$$

Equation (20) indicates that the parameter $\beta(k)$ can be calculated by simple matrix inversion of an upper triangular matrix R^{-1} and by simple matrix multiplication with Q^T instead

of calculating $P^{-1} = R^{-1}Q^{-1}$. The methodology is not only optimal and numerically stable but also provides the tracking of varying parameters.

C. Linear Discriminate Analysis for Classification

For classification purposes, LDA is used. The adaptive autoregressive parameters that were obtained for every sampling time instant at each EEG channel are kept in a vector array. The produced vector V contains the AAR parameters of all EEG channels given as

$$V(i) = [\beta(i)^{Chn1}; \beta(i)^{Chn2}; \dots; \beta(i)^{ChnN}], \quad (21)$$

where $\beta(i)^{Chn1}; \beta(i)^{Chn2}; \dots; \beta(i)^{ChnN}$ represent the parameters which have been estimated for EEG time series measured at channels Chn1, Chn2, ..., ChnN, respectively, by the adaptive algorithms. The classification decision is made by calculating weight vector d and distance $dis(i)$ by

$$val(i) = d - V(i)d_0. \quad (22)$$

Here $val(i)$ is the output of LDA classifier for i^{th} trial. The 75% of the trials have been used for training the classifier and the remaining are used for testing the classification accuracy. The trials for which $val(i)$ has a value greater than zero are classified as those task periods which indicate right hand movement imagination. The trials for which $val(i)$ has a value lesser than zero are categorized as task periods belonging to left hand movement imagination.

D. EEG Data Acquisition

The proposed QR decomposition based methodology is validated by using datasets from motor imagery experiments conducted at PIEAS using Emotiv Epoch head-set [17] and a publicly published Graz dataset from the Institute for Biomedical Engineering, University of Technology Graz [10]. Datasets are acquired using similar experimental protocol. Here is the experimental paradigm from which the datasets are obtained.

Experimental Paradigm. Subjects are instructed about the requisite behavior during experimental trial by a visual stimulus. The experiments consist of multiple trials, which last for 10 seconds each. In every trial, a black colored fixation cross appears at the middle of computer screen (in case of PIEAS data) or position of a feedback bar (in case of Graz data) which lasts for four seconds. During the fifth second of each trial, the subject is commanded for left or right hand movement imagination with an appropriately directed arrow. The screen is then left blank next five seconds of trial. During these five seconds, the EEG signals are acquired from corresponding channels. Trials are randomized during the entire experiment to avoid subject's adaptability. Datasets in PIEAS are recorded from five healthy subjects with average age of 23.5 years and are denoted by ‘A’. The Graz dataset is acquired with a normal 25 year female subject. Motor imagery signals are related with

motor cortex area in human brain. Thus, the corresponding locations for signal acquisition are from electrodes present at the locations C3, Cz and C4 according to the international 10-20 system of electrodes. The signals obtained from three bipolar EEG channels using EEG head-sets with Ag/AgCl electrodes are used for analysis. Sampling frequency is set to 128Hz. The considered datasets are filtered with band pass filter having cutoffs at 0.5Hz and 30Hz, respectively.

III. RESULTS

Validation of the proposed EEG signal decoding methodology is performed by a comparative analysis. EEG features are extracted from four adaptive algorithms, i.e., the presented QR-RLS algorithm with forgetting factor ‘ w ’, LMS, RLS and the conventional QR-RLS algorithm. The obtained features are then classified with LDA. Performance of corresponding algorithm is determined by comparing classification accuracy acquired from it, in terms of percentage correct-trial classification (%CTC). A comparison of average %CTC values for Graz and PIEAS dataset with EEG processing algorithms is shown in Table 1. From table, it is evident that the proposed QR-RLS methodology with forgetting factor provides better classification results as compared to other EEG processing algorithms.

IV. DISCUSSION

Brain activity inside a human brain depends on bunch of neurons that generate spikes and then transfer it to their neighboring bunches. Neural firing is a well-accepted time variant phenomenon [18]. Therefore, an EEG data acquisition algorithm should be able to track the changes rapidly in time.

TABLE 1: COMPARISON OF AVERAGE VALUES OF THE PERCENTAGE CTC WITH THE FOUR ALGORITHMS.

		Percentage CTC (Avg)			
Adaptive Algorithms \ Datasets		LMS	RLS	QR-RLS	QR-RLS (w=0.95)
PIEAS BCI	A.1	55.12	55.27	60.42	65.50
	A.2	85.43	89.92	95.00	96.6
	A.3	54.67	55.56	58.75	63.89
	A.4	74.50	76.50	80.56	80.56
	A.5	57.92	59.03	62.08	67.78
GRAZ BCI		51.85	60.10	80.92	84.32

It is possible that during the trial when the subject is presented

a cue, the neural activity from various neurons changes to respond to the motor imagery task. Time variant decision making has been proven in many studies [19] and it allows the subject to perform the assigned task optimally – brain undergoes some dynamic adjustments in its activity parameters during the trial. It is infact possible that before the actual cue choice comes, the subject’s brain starts getting adapted to the previous pattern of trials. Though the randomized nature of the experiment, a saturation level might be achieved by the subject when they get adapted/learned to the given directions (left or right). Thus, entire brain activity needs to be considered as a time varying system.

QR-RLS algorithm with forgetting factor has capability of rapidly tracking the changing brain activity through the AAR parameters, besides providing numerical stability in their estimates as shown in Figures 1-2. As from Figure-1, with ‘ $w=1$ ’, the parameters are not tracked until the end of trial, whereas in Figure-2 with ‘ $w=0.95$ ’, parameters track the value of brain activity earlier. As a result, the classification accuracy is improved. Thus, it promises towards proficient processing of EEG signals and provides for a pathway for developing accurate, reliable and numerically stable EEG based BCI systems. An online brain imaging framework developed by utilizing the proposed approach can offer numerically stable and quicker medical diagnostic and rehabilitation facilities.

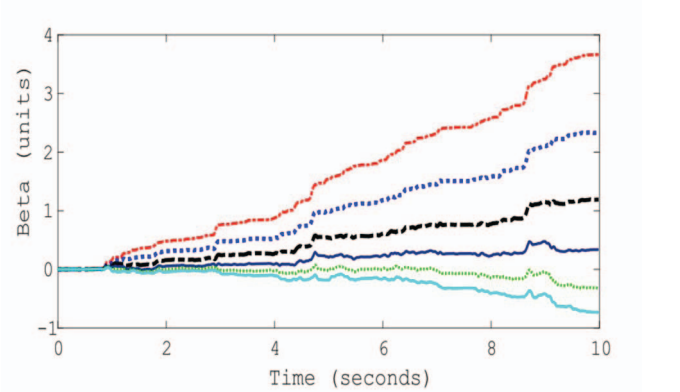


Fig. 1. β variation within a trial for QR-RLS with $w=1$.

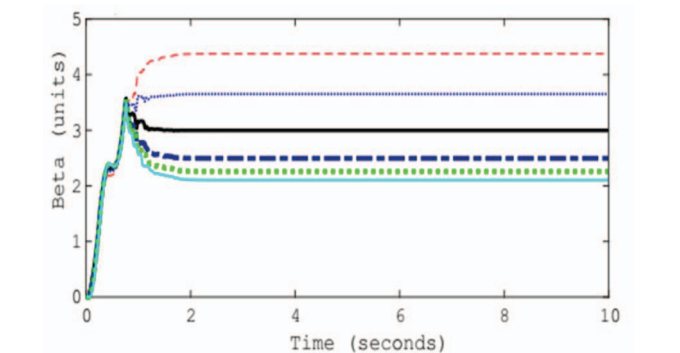


Fig. 2. β variation within a trial for QR-RLS with $w=0.95$.

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